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Collective Emergent Alignment in Multi-Agent LLMs

REMAP Quantifies Interaction-Induced Reasoning Gains; Gap-Aware Factual Messages Amplify Them by +3.1 F1 While Cutting Token Cost by 58%

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Multi-agent LLM systems often outperform any individual agent on reasoning tasks—but is this due to genuine peer interaction or simply to extra self-refinement compute? This paper introduces Collective Emergent Alignment (CEA) to measure the interaction-specific gain, and the REMAP framework to quantify it at the level of individual facts. Across seven model families and two multi-hop benchmarks, CEA is consistently +6.3 F1 points above matched self-refinement, and gap-aware factual communication amplifies this by a further +3.1 points while reducing communication tokens by 58%.

AT A GLANCE

Problem: It is unclear whether multi-agent LLM performance gains reflect genuine collective intelligence or just additional self-refinement compute.

Why it matters: Multi-agent LLM systems are increasingly deployed in production; understanding whether interaction actually helps is essential for designing efficient systems.

Method: REMAP reference-entailment matrix measures per-fact alignment gains attributable to interaction; gap-aware factual messages route only missing facts between agents.

Result: CEA is +6.3 F1 over matched self-refinement; gap-aware communication adds +3.1 F1 and cuts tokens by 58%.

Evidence: 7 model families × 2 multi-hop QA benchmarks (MuSiQue, 2WikiMultiHopQA); group sizes 2, 4, 8.

The Big Picture

In a multi-agent LLM system, several models communicate over multiple rounds to jointly arrive at an answer. The intuition is that diverse agents will catch each other's errors and combine partial knowledge. Empirically, these systems often outperform single-model baselines. But this comparison is confounded: a multi-agent system at round k has performed more computation than a single agent that runs k self-refinement steps on its own. The improvement could be *collective*—arising from information transfer between agents— or *individual*—arising from additional self-refinement at matched compute.

This paper disentangles the two. It defines Collective Emergent Alignment (CEA) as the improvement in reference-grounded reasoning that arises specifically from inter-agent interaction, over and above what the same total compute would yield in independent self-refinement.

Why Existing Approaches Fall Short

Multi-agent vs. single-agent comparisons: Most prior work compares a multi-agent system directly to a single-agent baseline without controlling for compute. The multi-agent system always has more iterations and more tokens; any gain is ambiguous.

Aggregate metrics only: Existing evaluations report final F1 or accuracy but do not attribute performance gains to specific communication events. There is no way to tell which round or which message was responsible for a correct answer.

No formal model of collective gain: Prior work lacks a framework for defining what "collective" means in a measurable sense. Terms like "debate," "collaboration," and "consensus" are used loosely without formal grounding.

Core Mechanism

Reference-Entailment Matrix. REMAP constructs a binary matrix $M \in \{0,1\}^{A\times R\times F}$ where A is the number of agents, R the number of rounds, and F the number of atomic facts in the reference answer. $M_{a,r,f} = 1$ if fact f is entailed by agent a's response at round r. The matrix is populated using a pretrained natural-language inference model.

Collective Emergent Alignment. A fact f is attributed to CEA if:

- It first appears in some agent's response after a communication round (not at round 0), and
- It does not appear in the counterfactual trajectory where that agent self-refines without receiving peer messages.

The CEA score is the fraction of reference facts attributed to CEA under this definition.

Gap-Aware Factual Communication. Instead of exchanging full reasoning traces, each agent identifies which facts in its peer's last response are absent from or contradict its own response. It then routes a compact message containing only those "gap facts" to the peer. This is implemented with a lightweight LLM step that extracts

the gap-fact set from both responses.

Technical Formulation

Let $R_{a,r}$ denote agent a's response at round r, and let $E(R, f)$ be the entailment probability for fact f given response R. The binary matrix is:

$$M_{a,r,f} = \mathbf{1}[E(R_{a,r}, f) > \tau]$$

The Semantic Coverage of the collective at round r is:

$$\mathrm{SC}(r) = \frac{|\{f : \exists\, a,\ M_{a,r,f} = 1\}|}{|F|}$$

The CEA score is:

$$\mathrm{CEA} = \frac{1}{|F|} \sum_{f} \mathbf{1}[f \in F_{\text{collective}} \setminus F_{\text{individual}}]$$

where $F_{\text{collective}}$ are facts recovered by the multi-agent system and $F_{\text{individual}}$ are those recovered by the matched self-refinement baseline.

For gap-aware communication, the message from agent a to agent b at round r is:

$$\mathcal{G}^r_{a \to b} = \{f : M_{a,r,f} = 1,\ M_{b,r,f} = 0\}$$

and the message content is a natural-language summary of $\mathcal{G}^r_{a \to b}$ generated by a compact auxiliary LLM.

Learning or Inference Procedure

No model training is required for REMAP measurement. All LLMs are used in their pretrained, instruction-tuned states. The evaluation procedure is:

1. Initialize A agents with the same question.
2. Run R rounds of communication, alternating between standard full-trace and gap-aware factual modes.
3. Construct M using the NLI model after each round.
4. Compute SC, CEA, and per-round alignment gains.

The self-refinement counterfactual is obtained by re-running each agent independently for the same number of rounds, receiving no peer messages.

What the Guarantee Says

The paper provides no formal guarantee about CEA's optimality or about the gap-aware communication protocol. The theoretical contribution is the definition of CEA itself, which is grounded in the entailment matrix and admits a clean counterfactual interpretation. The gap-aware communication protocol is justified empirically: it consistently amplifies CEA and reduces token cost across all evaluated settings.

Experimental Findings

- **CEA is real:** Across 7 model families and both benchmarks, CEA averages +6.3 **F1 points** over matched self-refinement. All model families show positive CEA; none is negative.
- **Gap-aware communication:** Amplifies CEA by +3.1 **F1** while reducing communication tokens by **58%** compared to full-trace exchange.
- **Group size:** CEA increases with group size (2 to 8 agents) but with diminishing returns; going from 2 to 4 agents yields +4.1 F1, while going from 4 to 8 yields only +1.8 F1.
- **Benchmarks:** MuSiQue (harder, requiring 4-hop reasoning) shows higher CEA (+8.1) than 2WikiMultiHopQA (+4.5), consistent with more room for collective benefit when individual models struggle.

Ablations and Interpretation

Communication timing: Communicating every round is better than every 2 rounds (+1.2 F1), but the gap narrows with gap-aware messages (which are shorter and can be communicated more frequently).

Message content: Full-trace exchange vs. gap-aware facts: full-trace exchange achieves 3.1% lower F1 and costs 58% more tokens. Gap-aware messages are both cheaper and better.

NLI threshold τ: CEA is robust to $\tau \in [0.6, 0.8]$; below 0.6, noisy entailment decisions inflate CEA artificially; above 0.8, the definition becomes too strict and CEA underestimates the true gain.

Dominance phenomenon: In 22% of episodes, one agent provides >70% of the CEA facts. Sequential communication reduces this to 9%, suggesting that synchronous communication can lead to one agent dominating the group.

Reference

Boxuan Wang, Tianxiang Sun, Linyang Li, Xipeng Qiu, Hang Yan. **Collective Emergent Alignment: Measuring and Amplifying Interaction-Induced Reasoning Alignment in Multi-Agent LLMs.** arXiv:2610.34836. 2026. https://arxiv.org/abs/2610.34836